A Coarse-to-Fine Hyperspectral Target Detection Method Based on Low-Rank Tensor Decomposition

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***Abstract*— To solve the problem of low target detection accuracy caused by the related quantities such as background, target, and noise contained in hyperspectral images (HSIs), considering the use of the spatial spectrum and spectral characteristics while increasing the degree of discrimination between target and background, a coarse-to-fine HSI target detection algorithm based on low-rank tensor decomposition (HTDLTD) is proposed. The HTD based on low-rank sparse decomposition mainly decomposes HSIs in spectral dimension, which does not make full use of the spatial information of HSIs, resulting in low detection accuracy. In order to solve this problem, in view of the fact that the hyperspectral third-order tensor can describe the spatial information and spectral information of HSIs equally, the HTD method based on low-rank tensor decomposition (LRTD) is proposed to extract pure background information. Then, in order to solve the problem of low detection accuracy in the case of low target and background discrimination, the rough target detection method based on max over (SMF-MAX) target detection method is proposed to perform rough detection on the original HSI to obtain rough detection results. Finally, in order to further improve the performance of target detection, the fine target detection method based on spectral distance is proposed. By calculating the spectral distance between the original HSI and the synthesized HSI, the final reconstructed target detection result is obtained. Experimental results on three datasets show that the proposed HTDLTD exceeds eight state-of-the-art target detection methods used for comparison.**

***Index Terms*— Hyperspectral images (HSIs), hyperspectral target detection (HTD), low-rank tensor decomposition (LRTD).**

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I. Introduction

HYPERSPECTRAL remote sensing, with the abundant spectral information provided by the hyperspectral data, plays a critical role in a wide spectrum of applications [1], [2], [3]. Target detection focuses on separating the target of interest from background interference and is one of the most important application directions for hyperspectral images (HSIs) [4], [5], [6]. Based on the availability of the prior information of the target, target detection plays a unique advantage in such tasks of finding specific objects [7]. Theoretically, target detection can be considered as a binary hypothesis testing problem where each test pixel is labeled as a target (target present) or background (target absent) [8]. As a vital part of hyperspectral remote sensing, the HTD has been involved in a number of applications for both military and civilization [9], [10], [11], [12], [13].

Since the 1990s, some researchers have applied the signal-processing theory to real HSI processing and established a basic design framework for target detection in HSIs [1]. Traditionally, the spectral difference between the target and background is exploited to detect targets. Constrained energy minimization (CEM) [21] is a classical HSI target detection algorithm. The algorithm only needs to know a target spectrum of interest to achieve target detection in an unknown background environment. However, the accuracy and stability of the CEM algorithm often encounter technical bottlenecks in the complex scenarios of practical applications. Many extensions based on CEM have been investigated for hyperspectral target detection (HTD) [22], [23], [24]. The squared CEM is defined as the normalized generalized signal-to-noise ratio (NGR-SNR) [25]. This method separates the subpixel/mixed-pixel target from the background (BKG) as a single target pixel, so that the target can be detected by detecting the abundance fraction of the target.

In addition to the above detectors, there are other detectors that are designed based on mixed models. The adaptive coherence/cosine estimator (ACE) [26] and the spectral matched filter (SMF) [27] use the properties of the correlation or covariance matrix of the image under test [28], [29], [30]. The derivation of the decision functions is based on both the statistical theory and subspace projection. Based on these classical algorithms, many improved methods were developed, like the subpixel SMF (SPSMF) [31] and the SMF and max over (SMF-MAX) target detection method [32]. The SPSMF is the generalized likelihood ratio test (GLRT) derived algorithm

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under a hypothesis model that should be more suitable for hyperspectral imagery than the SMF models. Taking the SMF model as a starting point, the SPSMF removes the scale term under the null hypothesis and allows for a nonzero background mean. The traditional SMF has a limitation in that it assumes that the target signal is added to an existing background signal versus replacing the background signal with target information. To alleviate some of this effect, taking the max over targets of the SMF results in a more robust detection statistic often referred to as SMF-MAX.

Under reasonable assumptions, the above traditional algorithms perform well and exhibit robustness [33]. If the description of the model for data in real scenes is not accurate enough [34], that is, the model does not match the real HSI data distribution well, their detection performance may decrease significantly [35]. Moreover, most classic algorithms have neglected the spatial information of HSI which may be crucial for detection.

With decades of development, kinds of deep-learning models have been applied for HSI target detection. Deep-learning-based methods are generally regarded to yield higher detection accuracy than those traditional shallow feature extractors [36]. Li et al. [37] proposed a transferred learning detector for HTD based on the convolutional neural network (CNN), called CNNTD. It uses a labeled reference dataset to train a CNN, which is then used for HTD. To further boost the denoising performance, many researchers proposed a variety of HSI mixed noise removal models based on the framework of the CNN [38], [39], [40], [41], which has the powerful nonlinear fitting and feature expression ability. For example, Chang et al. [38] proposed a CNN-based method by modeling the spatial and spectral features by the dilated convolution and multichannel filters, respectively. However, deep-learning-based methods often suffer from insufficient labeled samples.

With no need for any explicit assumption about the spectral distribution of an HSI, the representation-based detectors are an important migration in the field of machine learning, which are an active research area of HTD [42], [43], [44], [45]. According to the compressed sensing theory, the sparse-representation-based detector (SRD) aims to build both the target and background dictionaries to represent the test pixels in HSIs. The combined sparse and collaborative-representation-based detector (CSCR) preserves the sparsity of the target, and the background components are collaboratively represented by its adjacent pixels [46], [47]. Xu et al. [48] proposed a tensor robust principal component analysis (RPCA)-based HSI reconstruction and anomaly detection method. In this method, the background and anomaly parts are separated effectively by the novel Mahalanobis distance regularization, and the detection accuracy is improved. In [49], an effective infrared small-target detection method based on three-order tensor creation and tucker decomposition (TCTD) was proposed. Tucker decomposition was applied to estimate and eliminate most of the background from three dimensions. Attouch et al. [50] proposed a new regularized tensor decomposition for low-rank tensor completion (LRTC). Besides these methods, Zhao et al. [51] proposed a novel robust low-rank and sparse tensor decomposition model (RLRST) for tensor completion. In order to effectively filter out sample noise and obtain accurate low-rank structures of tensor data. Inspired by these methods, we consider using low-rank tensor decomposition (LRTD) to reconstruct background information. From the viewpoint of general regularized tensor decomposition for LRTC, methods can be roughly categorized as that the LRTD theory can extract the pure background tensor of the HSI.

Xue et al. [52] proposed a dual sparsity approach (DSC) for HTD, which constructed the target and background dictionaries simultaneously and put a dual sparsity constraint on atoms in every dictionary to obtain purer dictionaries. However, most of these representation-based methods need to construct the target and background dictionaries simultaneously, which requires a lot of prior spectral information. Besides, some of them adopt the dual-window detection mode, and it is necessary to set appropriate sizes of the dual window according to the spatial information of the target to achieve the desired detection performance. When some signatures of materials are lacking in a learned dictionary, their performances may become worse. Given limited prior information, such detectors are restricted to some extent in practical applications [53]. However, the truncated rank estimation scheme with such algorithms is, in general, short of a clear physical meaning. Moreover, these methods can not fully explore the common characteristics of different bands and also ignore the influence of noise on HSIs.

To cope with the above problems, a coarse-to-fine HSI target detection method based on LRTD (HTDLTD) is proposed. The HTDLTD mainly aims to improve the target detection accuracy by making full use of the spatial and spectral information of an HSI and improving the discrimination between background and target pixels.

First, the LRTD method is used to extract the pure background tensor of the HSI. To be specific, the decomposed spatial tensor factor and the spectral matrix factor obtained by the LRTF-DFR method are subjected to Mode-3 product to synthesize the HSI image containing pure background information. Second, in order to solve the problem of low detection accuracy in the case of low target and background discrimination, the SMF-MAX target detection method is used to obtain coarse detection on the original HSI. The SMF-MAX shows good background suppression ability, so it can suppress the background of fine detection results to highlight the target, so as to get purer detection results. Finally, in order to obtain more accurate target detection results, targets are picked out by calculating the spectral distance between the original HSI and the reconstructed background HSI in the fine detection stage. So as to achieve the purpose of restraining the unexpected background spectrum and keeping the target spectrum unchanged, each pixel value of the reconstructed target detection image is multiplied by a nonnegative coefficient which is based on the output of the coarse detection to obtain the final target detection result.

The main contributions are summarized as follows.

1) First, to fully exploit the spatial-spectral characteristics of HSIs without destroying the spatial-spectral structure, we propose a method based on LRTD to extract background information. Different from traditional LRTD,

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this method extracts pure background information based on a two-factor regularized low-rank tensor, which can not only make full use of the spatial information of HSIs but also reduce noise while reconstructing the background.

2) Second, in order to solve the problem of low detection accuracy in the case that target and background cannot be well discriminated, the coarse target detection method based on SMF-MAX is proposed to obtain preliminary detection results on the original HSI. The coarse stage can focus on suppressing background pixels to the maximum extent, so as to improve the separability of the target and the background.
3) Finally, in order to further improve the performance of target detection, the fine target detection method based on spectral distance is proposed. In the fine detection stage, the spectral distance between the synthesized HSI reconstructed by LRTF-DFR and the original HSI is calculated pixel by pixel to obtain the reconstructed target detection result. To further increase the distinction between background pixels and targets, the coarse detection results are used to further constrain the background of the reconstructed target detection results, which can obtain more accurate final target detection results.

The overall structure of this article takes the form of five sections. Section II gives some notations and preliminary knowledge. Section III introduces the details of the proposed method. Experimental datasets, performance analysis, and computational complexity analysis are shown in Section IV. Finally, the conclusion is outlined in Section V.

II. Notations and Preliminaries

In this section, the definitions of related symbols and some basic theories are introduced, including tensor modes, the definitions of mode-n unfolding, and mode-n tensor-matrix product tensor operation. The tensor is the extension of the matrix, whose dimensionality is called the order. For example, a matrix is a second-order tensor, which is basically numbers arranged in a rectangle. In this article, we represent a scalar by a lowercase letter, e.g., y, a vector by a bold lowercase letter, e.g.,$\mathbf{y} \in \mathbb{R}^I$, a matrix by a bold capital letter, e.g., $\mathbf{Y} \in \mathbb{R}^{I\times J}$, and a tensor by a bold calligraphic letter, e.g., $\mathcal{Y} \in \mathbb{R}^{I_1\times I_2\times\cdots\times I_N}$. Tensor algebra is based on multilinear algebra. Some concepts and operations related to our work are listed as follows.

Definition 1 (Tensor Mode): Tensor mode refers to different dimensions of a tensor, e.g., a second-order tensor, matrix, consisting of two modes: column mode and row mode. A kth-order tensor has k modes.

Definition 2 (Mode-n Unfolding): Mode-n unfolding is also called mode-n matricization, which is the process of constructing a matrix containing all the mode-n vectors of a tensor. For example, a third-order tensor $\mathcal{Y} \in \mathbb{R}^{I\times J\times K}$ can be unfolded in three ways as mode-1, mode-2, and mode-3 vectors, respectively,

$$\begin{aligned}\left(\mathbf{Y}_{(1)}\right)_{(j-1)K+k,i} &= y_{ijk}\\ \left(\mathbf{Y}_{(2)}\right)_{(k-1)l+i,j} &= y_{ijk}\\ \left(\mathbf{Y}_{(3)}\right)_{(i-1)J+j,k} &= y_{ijk}.\end{aligned} \tag{1}$$

Definition 3 (Mode-n Tensor-Matrix Product): The mode-n tensor-matrix product of an $I_1 \times I_2 \times \cdots \times I_N$ tensor $\mathcal{X}$ and a $J \times I_k$ matrix A is an $I_1 \times I_2 \times \cdots \times I_N$ tensor denoted by $\mathcal{X} \times {}_k\mathbf{A}$ and satisfied

$$(\mathcal{X} \times {}_k\mathbf{A})_{i_1,\ldots,i_{k-1},j,i_{k+1},\ldots,i_N} = \sum_{i_k=1}^{n_k} x_{i_1,i_2,\ldots,i_N} \cdot a_{j,i_k}. \tag{2}$$

According to the abovementioned two definitions, we have

$$\mathcal{Y} = \mathcal{X} \times_k \mathbf{A} \Leftrightarrow \mathbf{Y}_{(k)} = \mathbf{A}\mathbf{X}_{(k)} \tag{3}$$

and

$$\begin{aligned}\mathcal{Y} &= \mathcal{X} \times {}_1\mathbf{A}_1 \times_2 \mathbf{A}_2 \cdots \times_N \mathbf{A}_N\\ &\Leftrightarrow \mathbf{Y}_{(k)} = \mathbf{A}_k\mathbf{X}_{(k)}(\mathbf{A}_N \otimes \cdots \otimes \mathbf{A}_{k+1} \otimes \mathbf{A}_{k-1} \otimes \cdots \otimes \mathbf{A}_1)^{\mathrm{T}}.\end{aligned} \tag{4}$$

III. Proposed Method

The flowchart of the proposed method is shown in Fig. 1. In the proposed framework, first, the LRTD method is used to extract the pure background tensor of the HSI. Second, in order to further improve the degree of distinction between the background and the target, the SMF-MAX target detection method is used to perform on the original HSI to obtain coarse detection results. Finally, to obtain more accurate target detection results, the spectral distance between the synthesized HSI image and the original HSI image is calculated pixel by pixel to obtain the reconstructed target detection result, and the coarse detection result is used to suppress the background of the reconstructed target detection result to obtain more accurate the final target detection result.

A. Low-Rank Tensor Decomposition

The LRTD theory is used to extract the pure background tensor of the HSI. The HTD based on low-rank sparse decomposition mainly decomposes HSIs in spectral dimension, which does not make full use of the spatial information of HSIs, resulting in low detection accuracy. In order to solve this problem, in view of the fact that the hyperspectral third-order tensor can describe the spatial information and spectral information of HSIs equally, the HTD method based on LRTD to extract pure background information is proposed. HSI has the same 3-D data structure as the tensor, and the background data of the HSI contains low-rank characteristics. Due to the influence of the external environment, especially the influence of additive noise, the HSI generated by the imaging spectrometer will inevitably produce image degradation. The noise that affects the image mainly includes Gaussian noise, impulse noise, stripe noise, impulse noise, and stripe noise. Therefore, the degraded HSI model is represented as

$$\mathcal{Y} = \mathcal{X} + \mathcal{N} + \mathcal{S}. \tag{5}$$

Let $\mathcal{Y} \in \mathbb{R}^{I\times J\times K}$ be an original HSI with noise, I and J are the length and width of the spatial spectrum, and

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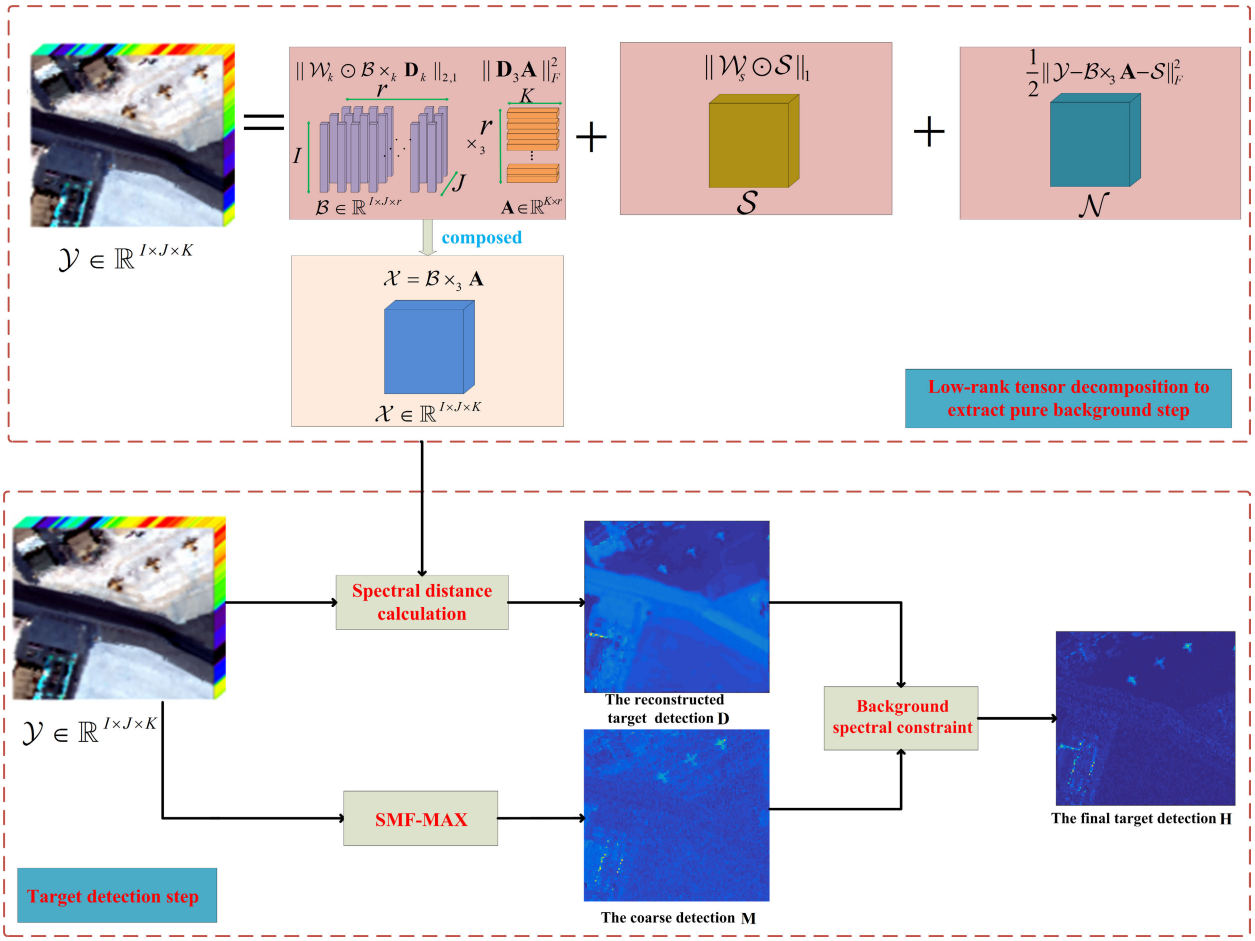


Fig. 1. Framework of the proposed method.

K is the dimension of the spectral dimension, using tensor decomposition theory, $\mathcal{X} \in \mathbb{R}^{I\times J\times K}$ is approximate pure HSI, $\mathcal{N}$ represents additive Gaussian noise, $\mathcal{S}$ indicates the additive sparse noise (salt and pepper noise, deadlines, and stripes).

There is a high degree of correlation between HSI bands, and there is a certain degree of redundancy. Therefore, the low-rank component can be represented by the strong correlation between the spectral band information. Mathematically, a generalized HSI restoration model can be formulated as

$$\min_{\mathcal{X},\mathcal{S}} \frac{1}{2}\|\mathcal{Y}-\mathcal{X}-\mathcal{S}\|_F^2 + \tau\mathcal{R}(\mathcal{X}) + \mu\|\mathcal{S}\|_1 \tag{6}$$

where τ and μ are regularization parameters and $\mathcal{R}(\mathcal{X})$ is the regularization term designed to exploit the HSI prior.

To extract pure background information and also reduce noise while reconstructing the background, we use a double-factor-regularized LRTF (LRTF-DFR) method for extracting low-rank background tensor $\mathcal{X}$

$$\min_{\mathbf{A},\mathcal{B},\mathcal{S}} \frac{1}{2}\|\mathcal{Y}-\mathcal{B}\times_3\mathbf{A}-\mathcal{S}\|_F^2 + \tau\sum_{k=1}^{2}\|\mathcal{W}_k\odot(\mathcal{B}\times_k\mathbf{D}_k)\|_{2,1} + \lambda\|\mathbf{D}_3\mathbf{A}\|_F^2 + \mu\|\mathcal{W}_s\odot\mathcal{S}\|_1 \tag{7}$$

where λ is a regularization parameter; $\mathcal{B} \in \mathbb{R}^{I\times J\times r}(r \ll K)$ denotes the spatial factor and $\mathbf{A} \in \mathbb{R}^{K\times r}$, $\mathbf{D}_k$ $(k = 1, 2, 3)$ are first-order difference matrices; $\mathcal{W}_k$ $(k = 1, 2)$ and $\mathcal{W}_s$ are weight tensors used to better promote the group sparsity [55] and the sparsity, respectively. Especially, all frontal slices of $\mathcal{W}_k$ are the same matrices, denoted as $\mathbf{W}_k$.

To solve the LRTF-DFR model, an efficient proximal alternating minimization (PAM)-based algorithm [57] was developed with a numerically guaranteed convergence, and the problem (5) can be solved by alternately updating as

$$\begin{cases} \text{Step 1: } \mathbf{A}^{l+1} = \underset{\mathbf{A}}{\operatorname{argmin}}\, f\left(\mathbf{A}, \mathcal{B}^l, \mathcal{S}^l\right) + \dfrac{\rho}{2}\left\|\mathbf{A}-\mathbf{A}^l\right\|_F^2 \\ \text{Step 2: } \mathcal{B}^{l+1} = \underset{\mathcal{B}}{\operatorname{argmin}}\, f\left(\mathbf{A}^{l+1}, \mathcal{B}, \mathcal{S}^l\right) + \dfrac{\rho}{2}\left\|\mathcal{B}-\mathcal{B}^l\right\|_F^2 \\ \text{Step 3: } \mathcal{S}^{l+1} = \underset{\mathcal{S}}{\operatorname{argmin}}\, f\left(\mathbf{A}^{l+1}, \mathcal{B}^{l+1}, \mathcal{S}\right) + \dfrac{\rho}{2}\left\|\mathcal{S}-\mathcal{S}^l\right\|_F^2 \end{cases} \tag{8}$$

where f $(\mathbf{A}, \mathcal{B}, \mathcal{S})$ is the objective function of (5) and $\rho > 0$ is the proximal parameter.

Fortunately, the spectral low rankness of HSIs can be represented as the mode-3 tensor-matrix product of a low-dimensional tensor and a low-dimensional matrix. Thus, the decomposed spatial tensor factor $\mathcal{B}$ and the spectral matrix factor $\mathbf{A}$ are multiplied by mode-3 to synthesize the HSI image $\mathcal{X}$ containing pure background pixels. Accordingly, the synthesized HSI $\mathcal{X}$ is represented as follows:

$$\mathcal{X} = \mathcal{B}\times_3\mathbf{A}. \tag{9}$$

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The synthesized HSI $\mathcal{X}$ will be used in the fine detection stage.

B. Coarse Detection

To solve the problem of low detection accuracy in the case of low target and background discrimination, the coarse target detection method based on SMF-MAX is proposed to perform coarse detection on the original HSI to obtain coarse detection results. The SMF-MAX is leveraged as a coarse detection approach to obtain the coarse detection map. The SMF-MAX algorithm is improved from the classical SMF algorithm, which can suppress background better so as to improve the target detection accuracy. First, the SMF can be derived by following the GLRT framework under the model assumptions:

$$\begin{aligned} H_0&: \quad \mathbf{y}_i = \mathbf{b} \\ H_1&: \quad \mathbf{y}_i = \alpha \mathbf{s} + \mathbf{b} \end{aligned} \tag{10}$$

where $\mathbf{b} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$ is the random background, σ is the average of background b, and $\mathbf{I}$ is an identity matrix. $\mathbf{s}$ is the target reference spectra with a scale factor α to account for illumination or mixing effects and $\mathbf{y}_i$ is the spectra under test.

The SMF uses a full covariance matrix and a potentially nonzero mean vector for the background $\mathbf{b} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. The result of maximizing the likelihood of the unknown parameters α, $\boldsymbol{\mu}$, and $\boldsymbol{\Sigma}$ is the detection statistic

$$r_{\mathrm{SMF}}(\mathbf{y}_i) = \frac{\left[\mathbf{s}^T \hat{\boldsymbol{\Sigma}}^{-1}(\mathbf{y}_i - \hat{\boldsymbol{\mu}})\right]^2}{\mathbf{s}^T \hat{\boldsymbol{\Sigma}}^{-1}\mathbf{s}} \tag{11}$$

where $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\Sigma}}$ are the sample background mean and covariance estimated from the image.

The SMF-MAX [32] algorithm is improved by taking the maximum value of the output result to get the rough detection result in $\mathbf{M} = [\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_{J\times K}]$

$$\mathbf{m}_i = \max(r_{\mathrm{SMF}}(\mathbf{y}_i)). \tag{12}$$

C. Fine Detection

In order to obtain more accurate target detection results, the original HSI $\mathcal{Y}$ is finely detected. Spectral distance is the difference in amplitude between two spectral curves expressed by calculating the distance between two spectral vectors in a K-dimensional space. Then, $\mathcal{X}$ and $\mathcal{Y}$ are compared pixel by pixel. The output $\mathbf{D}$ can be described as $\mathbf{D} = [\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_{I\times J}]$, where $\mathbf{d}_i$ is the spectral distance [59] between the $\mathbf{i}_{th}$ pixel in $\mathcal{X}$ and $\mathcal{Y}$, calculated by

$$\begin{aligned} \mathbf{d}_i &= \cos^{-1}\left(\frac{\mathbf{y}_i^T \mathbf{x}_i'}{(\|\mathbf{y}_i\|)\left(\left\|\mathbf{x}_i'\right\|\right)}\right) \\ &= \cos^{-1}\left(\frac{\mathbf{y}_i^T \mathbf{x}_i'}{\sqrt{\left(\mathbf{y}_i'\right)^T \mathbf{x}_i'}\sqrt{\left(\mathbf{y}_i^T \mathbf{x}_i\right)}}\right). \end{aligned} \tag{13}$$

The value of d_i which is smaller than the preset threshold δ should work as the background, the opposite is the target. Then, a distance map, containing distinguishable targets, is obtained.

Algorithm 1 The HTDLTD Algorithm

1. Double-factor-regularized low-rank tensor factorization
Input: The original HSI $\mathcal{Y} \in \mathbb{R}^{I\times J\times K}$, rank r, parameters $\tau, \lambda, \mu, \beta$, and $\rho = 0.1$.
Initialization: $l = 0, l_{\max} = 50, p_{\max} = 10, \mathbf{A}^0 =$ rand (n_3, r), $\mathcal{B}^0 =$ rand (n_1, n_2, r), and $\mathcal{S}^0 = 0$.
Update: $\mathbf{A}$, $\mathcal{B}$ and $\mathcal{S}$ via (8).
Synthesize the HSI $\mathcal{X}$ via (9).
Output: The synthesized HSI $\mathcal{X}$.
2. In the coarse detection stage
Input: The original HSI $\mathcal{Y} \in \mathbb{R}^{I\times J\times K}$.
The SMF-MAX method is used to perform coarse detection on the original hyperspectral image via (12).
Output: The coarse detection map $\mathbf{M}$.
3. In the fine detection stage
Input: The original HSI $\mathcal{Y} \in \mathbb{R}^{I\times J\times K}$, the coarse detection map $\mathbf{M}$, the synthesized HSI $\mathcal{X}$.
1): Obtain the reconstructed target detection result image $\mathbf{D}$ via (13).
2): Obtain the final target detection result $\mathbf{H}$ via (14).
Output: The coarse detection map $\mathbf{H}$.

To finally detect targets, each pixel value $\mathbf{d}_i$ is transformed by multiplying a nonnegative number $q(m_i)$ based on output of the coarse detection $\mathbf{M}$ as

$$\mathbf{h}_i = q(m_i)\mathbf{d}_i. \tag{14}$$

The output $\mathbf{H}$ can be described as $\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_{I\times J}]$, the nonlinear function $q(x)$ [60] is used to suppress the undesired background spectra while keeping the target spectra unchanged. Considering that the target pixels in the distance map are much greater than the background, $q(x)$ should be able to maintain large values and remove small ones. Here, the nonlinear suppression function is defined as

$$q(x) = \begin{cases} 1 - e^{-\epsilon x}, & x \geq 0 \\ 0, & x < 0 \end{cases} \tag{15}$$

where ϵ is a positive parameter to adjust the suppression performance. Finally, the detection result $\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_{I\times J}]$ is obtained.

By summarizing the aforementioned solving process, the pseudocode of HTDLTD is described in Algorithm 1.

IV. Experiments

In this section, experiments are conducted to validate the effectiveness of the proposed method. The details of three HSI datasets used in the experiments are introduced first. Next, the selected evaluation criteria and the relevant parameter settings of the experiment are given. Finally, the experimental results of the proposed method and the compared methods are analyzed.

A. Datasets Description

Our experiments are performed on three datasets, which are described as follows.

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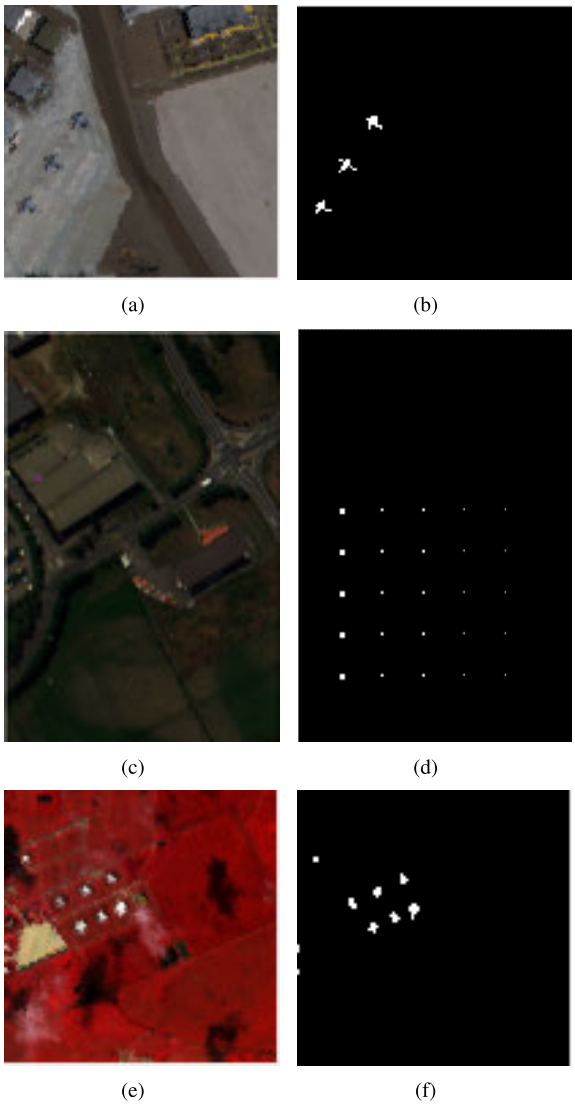


Fig. 2. False-color images: (a) R1, (c) R2, and (e) R3. Ground-truth maps: (b) R1, (d) R2, and (f) R3.

1) San Diego Dataset: The first hyperspectral dataset (hereinafter referred to as R1) is acquired by the airborne visible infrared imaging spectrometer (AVIRIS) sensor, covering the San Diego airport area, USA [60]. The spatial size is of text 120 × 120 pixels and it includes 224 spectral channels in wavelengths ranging from 400 to 2500 nm. The spatial resolution is approximately 3.5 m. In the experiments, a total of 189 bands were used after removing water-absorption and low SNR bands. The false color image of the R1 HSI dataset is shown in Fig. 2(a), and its corresponding ground-truth map is shown in Fig. 2(b).

2) Synthetic HSI Datasets: The second synthetic HSI dataset (hereinafter referred to as R2) was collected by the ROSIS sensor, over Pavia University, Northern Italy. The spatial resolution is 1.3 m, and the spectral resolution is 4 nm. The R2 dataset consists of 300 × 200 pixels and the number of its spectral bands is 103. The false color image of the R2 HSI dataset is shown in Fig. 2(c), and its corresponding ground-truth map is shown in Fig. 2(d). It can be found that there are 25 targets in total, and the sizes of these targets are 2 × 1, 1 × 2, 2 × 2, and 3 × 2, respectively.

3) Urban Dataset: The third HSI dataset (hereinafter referred to as R3) is collected over the urban of Texas Coast, USA [62]. It is also captured by the AVIRIS sensor, and its spatial resolution is 17.2 m. There are 204 spectral bands in this image, whose wavelengths range from 400 to 2500 nm. The image size of the HSI data is 100 × 100 pixels, and are chosen as targets. The false color image of the R3 HSI dataset is shown in Fig. 2(e), and its corresponding ground-truth map is shown in Fig. 2(f).

B. Evaluation Criteria and Experimental Settings

In this section, the evaluation criterion, compared methods, and parameter settings related to experiments are introduced as follows.

1) Evaluation Criteria: To evaluate the performance of the detection algorithms fairly and reasonably, the quantitative indicators used in our experiments are [63], i.e., the receiver operating characteristic (ROC), the area under the curve (AUC) metric and the separability mapbox.

2) Compared Methods: To verify the performance of the proposed method, eight HTD methods which are commonly used as compared methods in many studies are selected for complete comparison. The compared methods include the ACE method, CEM method, SMF method, SPSMF method, SMF-MAX method, NGR-SNR method, CNNTD method, and DSC method. These methods are selected as competitors by considering the diversity and popularity comprehensively. Among them, some classical algorithms are used as comparison algorithms, such as the ACE method, the CEM method, and the SMF method. In order to be more persuasive, this article also adopts two popular methods based on deep learning and machine learning in recent years as comparison algorithms, which are the CNNTD method and the DSC method.

3) Parameter Settings: In the HTDLTD model, there are several parameters that need to be carefully identified. The HTDLTD method involves six important parameters: the regularization parameters τ, λ, and μ, the penalty parameter β, the rank r, and adjust the positive parameter ϵ of the suppression performance. The parameters associated with the LRTF-DFR are set according to [51], and the optimal parameters settings for the HTDLTD method are shown in Table I. All parameters involved in the compared methods are carefully adjusted based on the authors' suggestions in their papers to obtain optimal performance. The gray values of HSIs are normalized into the interval [0, 1] band-by-band. For each dataset, a target pixel is randomly selected as the target priori [64], [65], [66]. All experiments are implemented by using MATLAB (R2015b) on Windows 7 with an Intel Core i5-5200U CPU and 4 GB of RAM.

C. Experimental Results and Analysis

In this section, first, the performance of the proposed HTDLTD method is evaluated and compared with eight widely used target detection algorithms, i.e., ACE [26], CEM [21], SMF [27], SPSMF [31], SMF-MAX [32], NGR-SNR [25],

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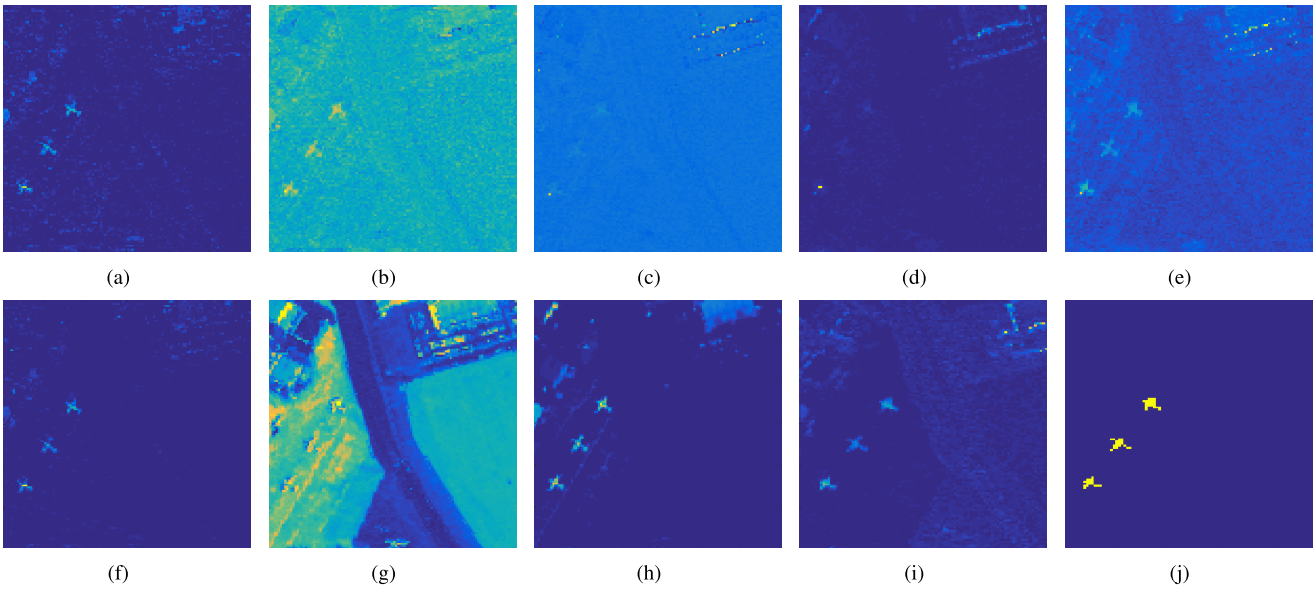


Fig. 3. Two-dimensional plots of detection results for R1. (a) ACE. (b) CEM. (c) SMF. (d) SPSMF. (e) SMF-MAX. (f) NGR-SNR. (g) CNNTD. (h) DSC. (i) HTDLTD. (j) Ground-truth.

TABLE I
OPTIMAL PARAMETERS SETTINGS FOR THE HTDLTD METHOD

Parameters	Datasets		
	dataset I	dataset II	dataset III
τ	0.2	0.2	0.2
λ	0.01	0.01	0.01
μ	0.03	0.05	0.04
β	15000	15000	15000
r	12	10	12
ϵ	10	10	8

TABLE II
AUC SCORES OF NINE DIFFERENT METHODS ON THREE DATASETS

Methods	AUC		
	R1	R2	R3
ACE [26]	0.9769	0.9800	0.9312
CEM [21]	0.9899	0.9883	0.9927
SMF [27]	0.9800	0.9625	0.9401
SPSMF [31]	0.9637	0.9605	0.9914
SMF-MAX [32]	0.9899	0.9786	0.8716
NGR-SNR [25]	0.9866	0.9872	0.8789
CNNTD [37]	0.9293	0.9082	0.9139
DSC [49]	0.9765	0.9230	0.9898
HTDLTD	**0.9976**	**0.9992**	**0.9987**

CNNTD [37], and DSC [49]. Then, in order to further prove the effectiveness of the proposed algorithm, an ablation study is done. Finally, the parameter sensitivity and the computational efficiency are analyzed.

1) Effectiveness Assessment and Performance Comparison: To make an overall evaluation of the effect and performance of the HTDLTD, eight classical and state-of-the-art HTD algorithms are chosen for comparison, which include ACE [26], CEM [21], SMF [27], SPSMF [31], SMF-MAX [32], NGR-SNR [25], CNNTD [37], and DSC [49].

The visual detection maps in Figs. 3–5 reveal that the detection maps obtained by the proposed HTDLTD are closest to the ground-truth. The HTDLTD method can ensure that targets with different sizes and shapes are highlighted while suppressing the background of rich categories. Other algorithms are more likely to ignore targets, get distracted by the background, or lose the shape of the targets. In particular, in Fig. 3, HTDLTD can capture the shapes of aircraft of different sizes as well as their positions. However, SMF and SPSMF do not. Although ACE and CNNTD can also detect targets clearly, they are greatly influenced by the background. CEM and SMF-MAX are barely affected by its background in each dataset. Nevertheless, in Fig. 4, the CEM method has poor robustness to overcome noise and background, and in Fig. 5, the SMF-MAX can not completely detect targets. The NGR-SNR has better robustness to overcome noise, but in Fig. 5, the NGR-SNR almost lost the targets. The CNNTD is also susceptible to background interference. As discussed above, it can be concluded that the HTDLTD method is competitive and robust.

Fig. 6 plots their ROC(s), and it can be clearly seen that the ROC curve of the proposed HTDLTD always remains over those compared methods. In order to quantitatively compare the performance of all algorithms, Table II listed the AUC values of all these algorithms. As shown in Table II, the HTDLTD significantly improves the detection accuracy. The HTDLTD has the best performance among these algorithms and the AUC values detected on the three datasets are 0.9976, 0.9992, and 0.9987, respectively. To further visualize the separation of background and targets, the target-background separability map on three HSIs is shown in Fig. 7. The red and green boxes in Fig. 7 represent the distribution of the targets' detection values and the background's detection values, respectively. Their positions illustrate the separability of targets and backgrounds. For the R1 dataset shown in Fig. 7(a), the gaps between targets and background for the proposed HTDLTD method and SMF-MAX method are bigger than the other methods. However, the upper whisker of the background

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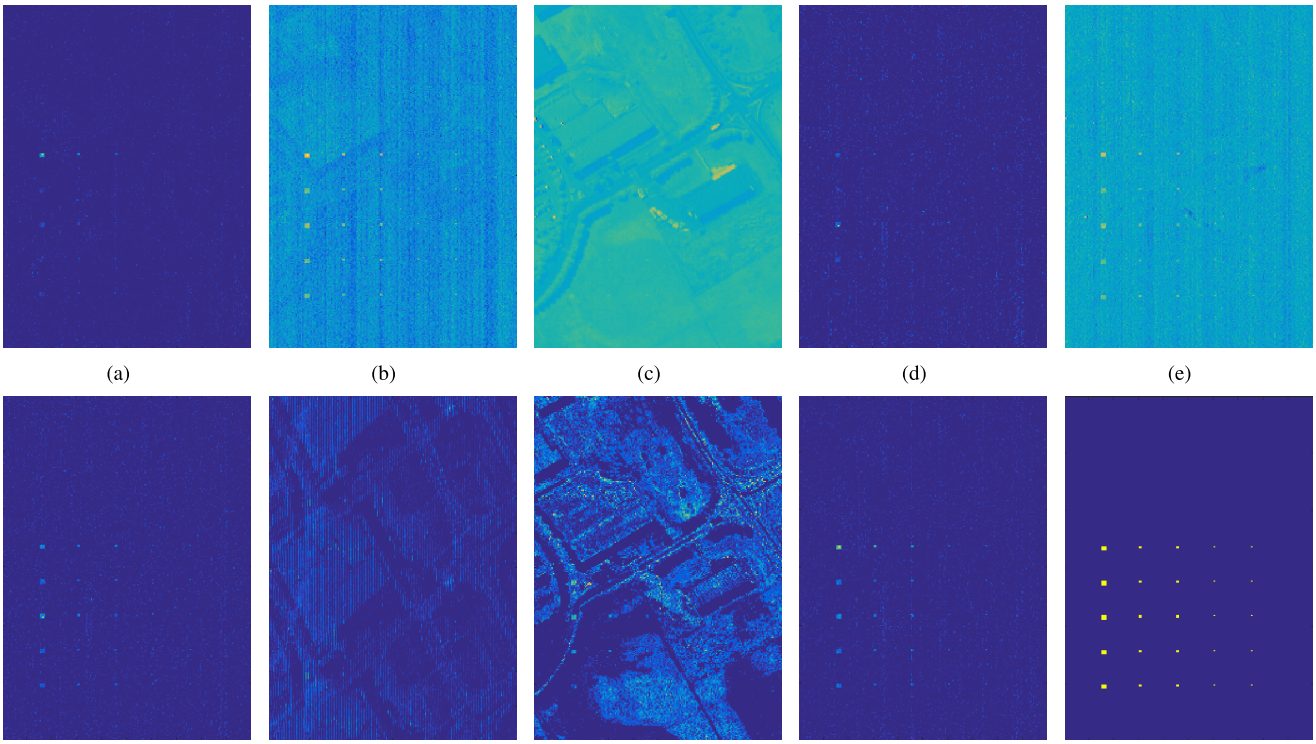


Fig. 4. Two-dimensional plots of detection results for R2. (a) ACE. (b) CEM. (c) SMF. (d) SPSMF. (e) SMF-MAX. (f) NGR-SNR. (g) CNNTD. (h) DSC. (i) HTDLTD. (j) Ground-truth.

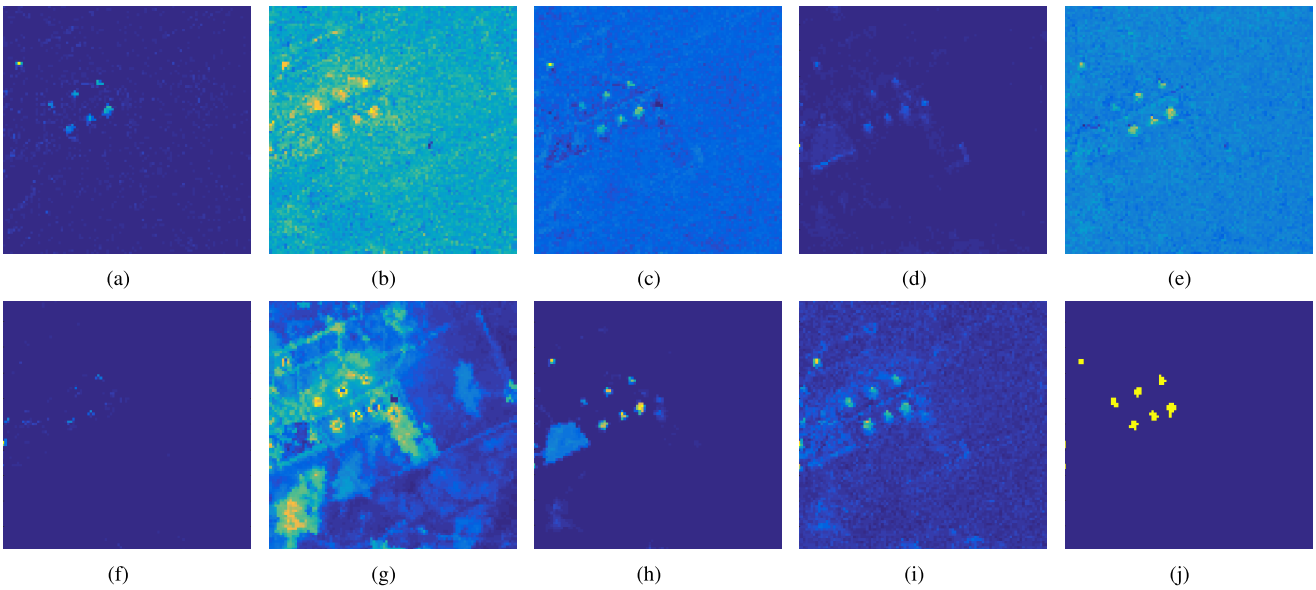


Fig. 5. Two-dimensional plots of detection results for R3. (a) ACE. (b) CEM. (c) SMF. (d) SPSMF. (e) SMF-MAX. (f) NGR-SNR. (g) CNNTD. (h) DSC. (i) HTDLTD. (j) Ground-truth.

box for the HTDLTD method is lower than SMF-MAX's, which means that the proposed HTDLTD method suppresses the background more effectively than SMF-MAX method. For the R1, R2, and R3 as shown in Fig. 7(a)–(c), NGR-SNR and DSC can hardly see the background, and the SPSMF method of the background and targets can not be separated. The target-background separability map indicates that the background and targets can be separated well by the proposed HTDLTD method. It can be easy to draw similar conclusions that the proposed HTDLTD produces the best performance among these compared methods. The 2-D visualized detection maps of all nine methods on three datasets are revealed in Fig. 7. As shown in Fig. 7, the HTDLTD can detect more target pixels. In addition, the HTDLTD can effectively detect the targets and retain the shape of the target, which may be due to the effectiveness of the coarse-to-fine strategy. Moreover,

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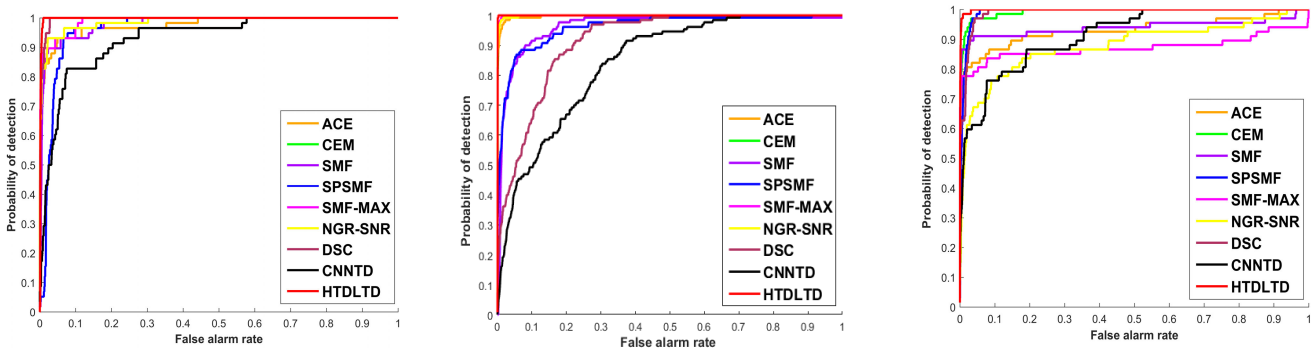


Fig. 6. ROC curves of nine target detection methods using three HSIs. (a) R1. (b) R2. (c) R3.

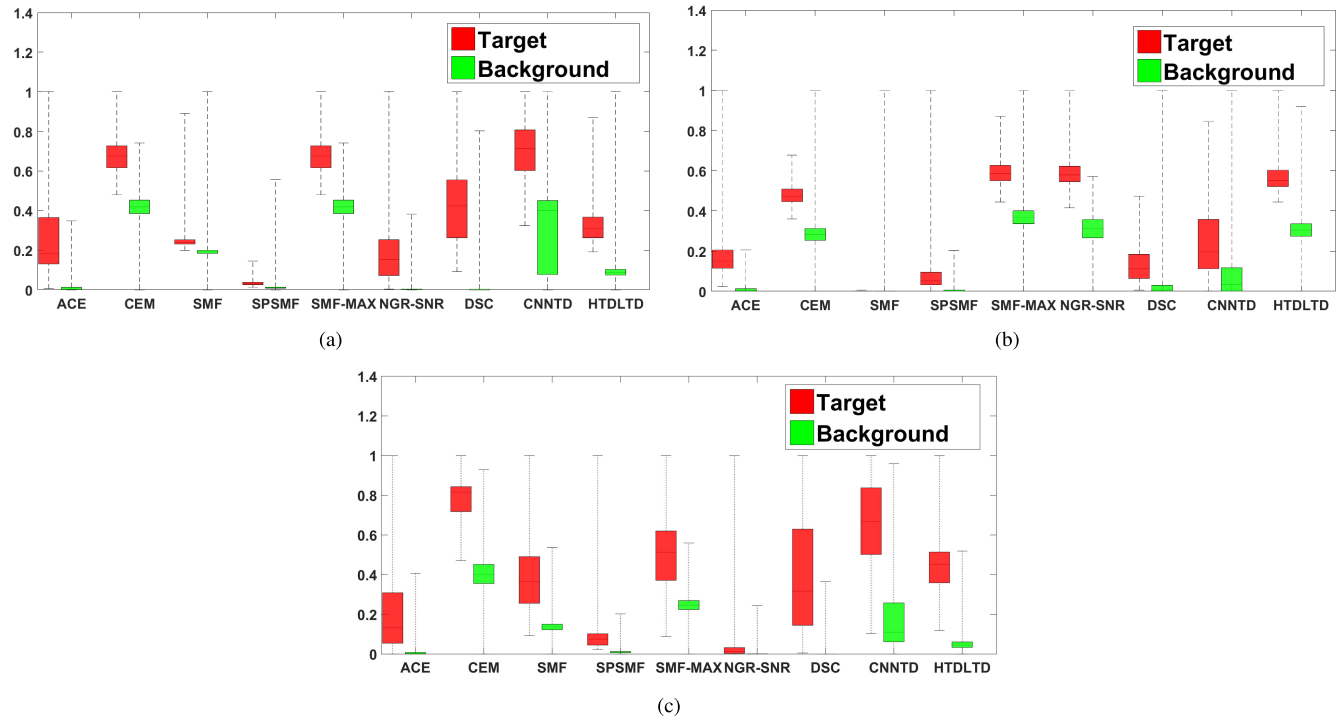


Fig. 7. Target-background separability map for the compared methods on three HSIs. (a) R1. (b) R2. (c) R3.

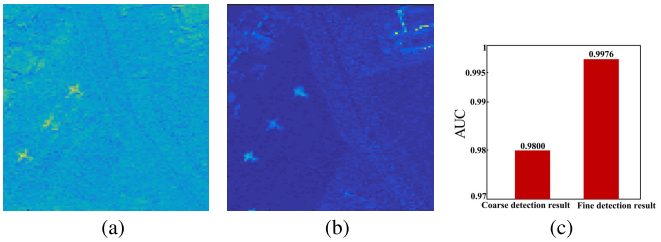


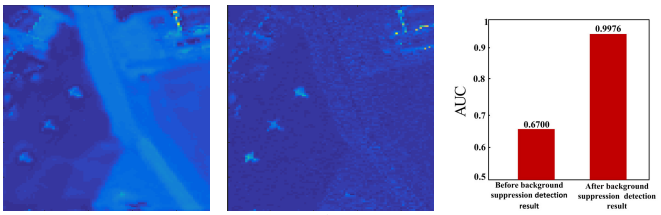
Fig. 8. Two-dimensional plots and AUC value comparison on R1. (a) Coarse detection map. (b) Fine detection map. (c) AUC value comparison between the coarse detection result and the fine detection result.

HTDLTD can better suppress background and noise, so that targets can be better detected.

2) Ablation Study: To justify the benefits of the proposed HTDLTD method, the following ablation studies are conducted on the dataset R1 in Fig. 8: Fig. 8(a) is the coarse detection map means that HSIs are directly fed into the SMX-MAX detector to use coarse detection without any work. Fig. 8(b) is the fine detection map after the fine detection strategy processing. Fig. 8(c) shows the AUC value comparison between the coarse detection result and the fine detection. Therefore, the coarse-to-fine HTD method is further improved by 1.76%, which indicates that the proposed method has a positive influence on improving the degree of distinction between the background and the target.

An HSI generally has a very complicated background including many unknown and unidentified landcovers other than noise. As a result, one of the major challenging issues arising in HTD is how to cope with background effects. In order to show the effectiveness of the method we use to suppress the background in the fine detection stage, we have verified it from two aspects in Fig. 9: the visual detection result map and the AUC value. Fig. 9(a) is the reconstructed

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(a) (b) (c)

Fig. 9. Two-dimensional plots and AUC value comparison on R1. (a) Reconstructed detection map. (b) Fine detection map. (c) AUC value comparison between the reconstructed detection result and the fine detection result.

TABLE III
AUC VALUES UNDER DIFFERENT ϵ FOR THREE DATASETS

ϵ	Datasets		
	R1	R2	R3
6	0.9798	0.9889	0.9899
8	0.9887	**0.9992**	0.9901
10	**0.9976**	0.9885	**0.9987**
12	0.9901	0.9768	0.9830
14	0.9866	0.9800	0.9651

target detection result map which is obtained by using spectral distance. Fig. 9(b) is the fine detection result map. Fig. 9(c) shows the AUC value comparison between the reconstructed target detection result and the fine detection result. That is to say, when the background spectral constraint is introduced, the performance of the system improves by an average of 3.149%. Commonly, although the use of spectral distance can effectively detect the target, the false alarm rate is also very high. By further using background suppression techniques, the performance of the proposed method can be enhanced.

3) Parameter Sensitivity Analysis: As shown in Algorithm 1, the HTDLTD method involves six important parameters: the regularization parameters τ, λ, and μ, the penalty parameter β, the rank r, and adjust the positive parameter ϵ of the suppression performance.

The parameter ϵ in target detection enables the nonlinear transformation will not hurt any target spectra while suppressing the background, which should be determined correctly. The selected ϵ range is from 6 to 14, and the AUC values with different ϵ are analyzed. The effect of ϵ is analyzed in Table III. When ϵ is set as 10, 10, and 8 in three datasets, respectively, our method has the best detection capacity. The rank r mainly characterizes the spectral global correlation of HSIs. The sensitivity analysis of the rank r is presented in Fig. 10(a). The selected r range is from 4 to 26, and we analyze the mean of peak signal-to-noise rate (MPSNR) with different r. It is seen that the proposed HTDLTD exhibits stable and superior performance within a certain range of the rank r. Considering that larger r leads to higher computational complexity, we set r to be 12, 10, and 12 for the three datasets, respectively.

The parameter μ determines the weight of the sparse noise term. Fig. 10(b) shows the sensitivity analysis of μ. It is seen that the R1 and R2 have stable and superior performance when μ is larger than 0.03. For R3, the performance of HTDLTD is slightly sensitive to μ. The sensitivity of R3 to the μ parameter may be caused by the complexity of R3 image noise. The three datasets achieve higher MPSNR when μ is 0.03, 0.05, 0.04.

TABLE IV
RUNNING TIME OF NINE DIFFERENT METHODS ON THREE DATASETS

Methods	Running Time		
	R1	R2	R3
ACE [26]	0.250890	4.134903	0.058296
CEM [21]	0.064940	3.053001	0.069637
SMF [27]	0.161504	8.036316	0.028630
SPSMF [31]	0.534980	6.327964	0.319345
SMF-MAX [32]	0.892565	9.285415	0.047720
NGR-SNR [25]	27.620863	56.623299	27.618929
CNNTD [37]	105.2477	130.8798	64.7738
DSC [49]	59.247705	105.8178	51.4529
HTDLTD	**39.705414**	**65.247022**	**25.418923**

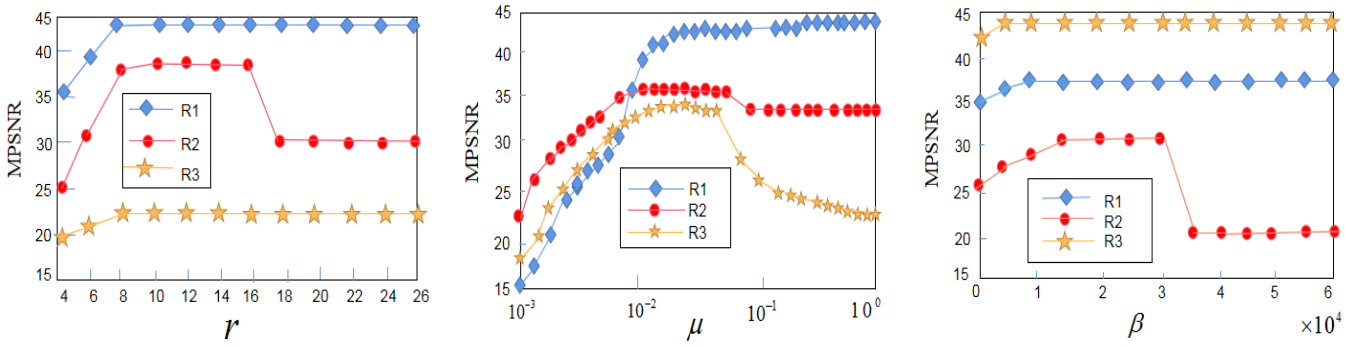
The sensitivity analysis of the penalty parameter β is shown in Fig. 10(c). It is obvious that the performance of the proposed HTDLTD is robust within a certain range of β. Similarly, through the analysis of parameters τ and λ on multiple experimental datasets, it is also found that the performance of the proposed HTDLTD is robust within a certain range of τ and λ. Therefore the three parameters of τ, λ, and β are set to the same value on the three datasets, 0.2, 0.01, and 15 000, respectively.

4) Convergence Analysis: Since the LRTD requires convergence and the weighted strategy are embedded in the PAM framework, it is necessary to prove the convergence of the developed algorithm in theory. A numerical example of relative change over iteration is used to prove the convergence. The "relative change" refers to the relative change of the image between the current iteration and the previous iteration in the LRTF-DFR. If the relative change gradually decreases with the increase of iteration times and finally stabilizes to a certain value, it shows that the method is convergent. Fig. 11 shows the relative changes of HSIs in each iteration and its previous iteration. The relative change ranges from 0 to 0.8 and the iteration ranges from 0 to 50. It can be observed from the figure that for all test datasets, the relative change value of the proposed HTDLTD algorithm decreases monotonically from 0.8, and gradually decreases to zero as the number of iterations increases to about 10, which can indicates the strong convergence of the proposed HTDLTD algorithm.

5) Computational Analysis: The computational complexity of the HTDLTD algorithm is analyzed on a degraded HSI $\mathcal{Y} \in \mathbb{R}^{I\times J\times K}$. The computational cost at each outer iteration mainly lies in the updating of $\mathbf{A}$, $\mathcal{B}and\mathcal{S}$. First, updating $\mathbf{A}$ via (8), which leads to $\mathcal{O}(rIJK + r^2K + rK\log(K))$ cost. Second, updating $\mathcal{B}$ via (8) needs $\mathcal{O}((r + I + J)IJK + r^2K + rIJ\log(IJ))$ cost. Third, the computational cost of updating $\mathcal{S}$ via (8) is $\mathcal{O}(IJK)$. Fourth, the computational cost of rough detection process via (12) is $\mathcal{O}(IJK)$. Finally, the computational cost of fine detection process via (13) is $\mathcal{O}(I^2Jr)$. In summary, the computational cost at each outer iteration of the HTDLTD algorithm is $\mathcal{O}(rK\log(K) + p((r + I + J)IJK + rIJ\log(IJ)))$.

Then, the computational efficiency is also a significant criterion for evaluating the performance of the HSI target detector. Table IV lists the running time of the HTDLTD method and compared methods. It can be found that the running time of HTDLTD is at a medium level in the running time of all comparison algorithms. In the proposed HTDLTD

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(a) (b) (c)

Fig. 10. Sensitivity analysis of the parameters on three datasets. (a) r. (b) μ. (c) β.

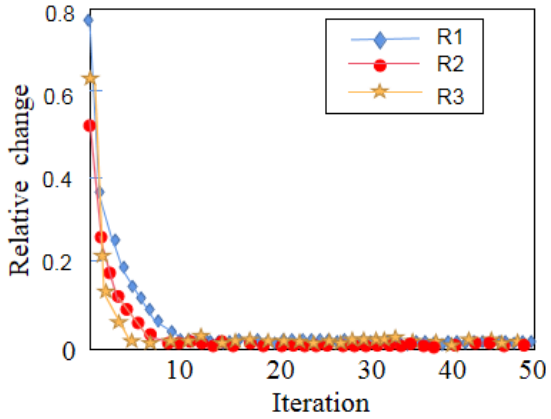


Fig. 11. Relative change curves with respect to the iteration number.

method, the LRTF-DFR algorithm which needs to be updated iteratively costs the most running time. However, considering the excellent detection accuracy of HTDLTD, the computational complexity of HTDLTD is acceptable.

V. Conclusion

In this article, a coarse-to-fine HTD method based on LRTD is proposed. The HTDLTD encourages the model to improve the HSI detection accuracy by enlarging the distinction between targets and backgrounds. The HSI has the same 3-D data structure as a tensor, and the background data of the HSI contains low-rank characteristics. Therefore, the LRTD method is used to extract the pure background tensor of the HSI. Then, in order to further improve the degree of distinction between the background and the target, the SMF-MAX target detection method is used to perform on the original HSI to obtain coarse detection results. Finally, to obtain more accurate target detection results, the spectral distance between the synthesized HSI image and the original HSI image is calculated pixel by pixel to obtain the reconstructed target detection result, so that the target information points in the image become prominent relative to the pure background information points. The coarse detection result is further used to suppress the background of the reconstructed target detection result to obtain a more accurate final target detection result. Through experimental comparison, the detection method which uses the spectral and spatial-spectral data structure characteristics of the HSI itself and introduces effectively the low-rank tensor can improve the detection accuracy of the classical target detection operator. Experimental results on three hyperspectral datasets show that the proposed HTDLTD is superior to the other target detection methods.

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